

Necessity, Reasoning, and Position: Complementarity-Aware Evidence Fusion for Multimodal Emotion-Cause Pair Extraction in Conversations

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Abstract

Multimodal emotion-cause pair extraction (MECPE) aims to identify, for every utterance that expresses an emotion in a conversation, the utterances that cause that emotion, using textual, acoustic, and visual signals together with speaker identity. Existing approaches model inter-utterance interaction but rarely ask *why* a candidate is predicted to be a cause: their decisions are not grounded in any explicit notion of necessity, and the strong positional regularity of causes is either ignored or allowed to dominate. We propose a model that couples a relation-aware multimodal encoder with three complementary sources of decision evidence: (i) perturbation-based *necessity evidence* that probes how much each candidate cause and each non-verbal modality contributes to a pair prediction; (ii) a *bounded positional prior* that exploits the locality of causes without letting position override semantics; and (iii) *reasoning evidence* distilled offline from a large language model (LLM) as text-grounded counterfactual judgements. A complementarity-aware gating network fuses these signals conditioned on model confidence, teacher reliability, and their agreement, so that each source compensates the other’s weakness; the LLM is queried only offline and never at inference. Experiments on the benchmark Emotion-Cause-in-Friends (ECF) corpus show that the proposed method improves the emotion-cause pair F_1 over strong baselines, and ablations confirm that every component contributes.

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1. Introduction

Understanding *why* a speaker feels a certain way is central to affective computing. Beyond recognizing that an utterance conveys an emotion, many applications—empathetic dialogue systems, opinion mining, and mental-health support—require identifying the events or utterances that *trigger* that emotion. The task of emotion-cause pair extraction (ECPE) formalizes this goal: given a document, jointly extract emotion utterances and their corresponding cause utterances as pairs (Xia and Ding, 2019). In conversations, emotions and their causes are conveyed not only through text but also through tone of voice and facial expression, which motivates the multimodal emotion-cause pair extraction (MECPE) task (Wang et al., 2023): for each emotion-bearing utterance in a multi-party conversation, find the utterances that cause it, using text, audio, and video.

MECPE is challenging for three intertwined reasons. *First*, the label space is quadratic in the number of utterances and highly imbalanced: only a small fraction of utterance pairs are genuine emotion-cause pairs. *Second*, a large share of causes are *self-contained*—the emotion is triggered by the event described *within* the emotion utterance itself, so the cause coincides with the emotion utterance. *Third*, causes exhibit strong but not absolute positional regularity: they tend to lie close to the emotion utterance, which is informative yet dangerous to over-trust.

Recent models substantially improve MECPE by modeling rich inter-utterance interaction with graph or transformer architectures and by injecting label constraints (Li et al., 2024). However, two limitations remain. (1) Their pair decisions are not grounded in any explicit notion of *necessity*: the model predicts a pair without ever asking whether removing the candidate cause would actually change the prediction, which makes decisions hard to calibrate and to explain. (2) External world knowledge—the kind of commonsense causal reasoning at which large language models (LLMs) excel—is not exploited, even though it is exactly the missing ingredient for distinguishing genuine triggers from mere topical co-occurrence. A naive remedy, querying an LLM per pair at inference, is impractical at the quadratic scale of MECPE and cannot perceive the non-verbal modalities.

We address these limitations with a model that augments a relation-aware multimodal encoder with *three complementary sources of decision evidence*. **(i) Necessity evidence** (Section 3.4) is a model-internal, perturbation-based signal: we replace the candidate cause—and, separately, each non-verbal modality of the cause utterance—with a learned, data-anchored neutral baseline, re-score the pair, and measure the resulting drop. A direct calibration loss anchors this signal to the pair labels. **(ii) A bounded positional prior** (Section 3.5) encodes relative position as a capped additive term on the pair score, with a regularizer that prevents position from overriding the semantic evidence. **(iii) Reasoning evidence** (Section 3.6) is distilled offline from an LLM: for hard pairs, a structured prompt elicits text-grounded *counterfactual* judgements (necessity, sufficiency, direction, spuriousness) together with a self-agreement reliability estimate; a lightweight student then predicts these scalars from the pair representation alone, so that no LLM is queried at inference.

The three signals are reconciled by a **complementarity-aware gating network** (Section 3.7) that conditions on the model’s confidence, the teacher’s predicted reliability, and the agreement between the model’s own necessity signal and the LLM’s necessity judgement. The design realizes an explicit division of labor: the LLM *compensates* the small model where the latter is uncertain, while the small model *leads* where the LLM is unreliable or disagrees. We further introduce a necessity-modulated agreement term so that consensus boosts the score only when both sources agree a pair *is* causal, preventing false-positive inflation when both agree it is *not*.

Our contributions are summarized as follows.

- We introduce a unified MECPE model that grounds pair decisions in explicit *necessity evidence*, obtained by faithful, pair-local perturbations of the cause utterance and of each non-verbal modality, and calibrated directly against the pair labels.
- We distill commonsense *counterfactual reasoning* from an LLM into a lightweight student with a dedicated treatment of self-contained causes, so that semantically informed external knowledge is available at inference *without* any LLM call.
- We propose a *complementarity-aware fusion* mechanism, together with a bounded positional prior and a necessity-modulated agreement term,

that lets each evidence source compensate the others’ weaknesses under explicit reliability and agreement conditioning.

- On the benchmark ECF corpus the method improves emotion-cause pair F_1 over strong baselines, and extensive ablations isolate the contribution of each component.

2. Related Work

Emotion cause analysis. Emotion cause extraction was first framed as detecting the clauses that cause a given emotion (Lee et al., 2010). Xia and Ding (2019) introduced emotion-cause pair extraction (ECPE), which removes the need for emotion annotations at test time by jointly extracting emotion and cause clauses as pairs. Subsequent work improved ECPE with two-dimensional representations (Ding et al., 2020) and effective inter-clause ranking (Wei et al., 2020). These methods operate on text alone and on monologue-style documents.

Multimodal emotion-cause pair extraction. Wang et al. (2023) extended the task to conversations and to multiple modalities, releasing the Emotion-Cause-in-Friends (ECF) corpus and establishing the MECPE task. Li et al. (2024) proposed a holistic-interaction model with label constraints that substantially advanced the state of the art. Our work builds on this line: we adopt a relation-aware multimodal encoder over the conversation graph, but depart from prior work by grounding pair decisions in explicit necessity evidence and by distilling external counterfactual reasoning, rather than relying on interaction modeling alone.

Graph and multimodal modeling of conversations. Graph neural networks are a natural fit for conversations, where utterances relate through speaker identity and reply structure (Ghosal et al., 2019). We use relation-specific attention in the spirit of relational GCNs (Schlichtkrull et al., 2018) and graph attention networks (Veličković et al., 2018). For multimodal fusion, prior work disentangles modality-invariant and modality-specific factors (Hazarika et al., 2020) and aligns modalities with cross-modal transformers (Tsai et al., 2019); we keep the fusion light and instead focus the modeling effort on causal-style evidence.

Interpretability and counterfactual reasoning. Perturbation-based attribution explains predictions by measuring sensitivity to input changes (Ribeiro et al., 2016; Lundberg and Lee, 2017). We repurpose this idea *inside* training: necessity evidence is a model-level signal that both regularizes and explains pair decisions, distinct from asserting causal relations in the world. Counterfactual reasoning has deep roots in causal inference (Pearl, 2009); we obtain weak, text-grounded counterfactual judgements by prompting an LLM, which is complementary to—not a substitute for—the model-internal perturbation signal.

LLMs and knowledge distillation. LLMs exhibit strong commonsense reasoning when prompted to reason step by step (Wei et al., 2022), but querying them at the quadratic scale of MECPE, and for non-verbal signals, is impractical. We therefore distill their reasoning offline into a small student (Hinton et al., 2015), restricting the LLM to text-grounded judgements and obtaining modality evidence from perturbations instead.

3. Method

We address MECPE in conversations, where the goal is to identify, for each utterance that expresses emotion, the utterances that cause it. Our model couples a relation-aware multimodal encoder with three complementary sources of decision evidence: necessity evidence (Section 3.4), a bounded positional prior (Section 3.5), and reasoning evidence distilled offline from an LLM (Section 3.6). The signals are combined by a gated fusion module (Section 3.7) and trained jointly (Section 3.8). Figure 1 gives an overview.

3.1. Task Formulation

A conversation is a sequence of N utterances $\mathcal{D} = \{u_1, \dots, u_N\}$. Each utterance carries textual, acoustic, and visual features together with a speaker identity, $u_i = \{x_i^t, x_i^a, x_i^v, s_i\}$. For an ordered pair (u_i, u_j) in which u_i is the candidate *emotion* utterance and u_j the candidate *cause* utterance, the model predicts

$$p_{ij} = P(y_{ij} = 1 \mid u_i, u_j, \mathcal{D}), \quad (1)$$

where $y_{ij} = 1$ iff u_j is a cause of the emotion expressed in u_i . Two auxiliary utterance-level heads predict an emotion probability $\hat{p}_i^e$ and a cause probability $\hat{p}_j^c$; these supply supervision and drive the candidate gating of Section 3.3.

Placeholder for the model overview figure: relation-aware encoder $\rightarrow$ soft candidate gate $\rightarrow \{\text{necessity evidence, positional prior, distilled reasoning evidence}\} \rightarrow \text{complementarity-aware fusion} \rightarrow \text{pair probability}.$

Figure 1: Overview of the proposed model. The LLM is used only offline to produce distillation targets; at inference the student supplies the reasoning evidence.

3.2. Relation-Aware Multimodal Encoder

Multimodal fusion. Each utterance is projected into a shared H -dimensional space by fusing its modalities and speaker identity,

$$h_i^0 = \text{Fuse}(x_i^t, x_i^a, x_i^v, s_i) \in \mathbb{R}^H. \quad (2)$$

Conversation graph. We build a heterogeneous graph over utterances with four relation types $\mathcal{R} = \{\text{self-loop, same-speaker, reply/adjacent, emotion-transition}\}$: *self-loop* ($i = j$); *same-speaker* ($s_i = s_j, i \neq j$); *reply/adjacent* ($|i - j| = 1$ or u_j is the reply target of u_i , with a fallback to the nearest preceding cross-speaker utterance); and *emotion-transition*, where structurally connected turns whose emotion labels differ are re-typed, taking priority over the structural off-diagonal types. The emotion-transition relation depends on emotion labels, which are gold during training and predicted at inference. To avoid a train-inference mismatch, the source label of each such edge is drawn by scheduled sampling (Bengio et al., 2015),

$$\ell_i = \begin{cases} \ell_i^{\text{gold}}, & \text{with probability } \varepsilon_t, \\ \ell_i^{\text{pred}}, & \text{with probability } 1 - \varepsilon_t, \end{cases} \quad \varepsilon_t : 1 \rightarrow 0, \quad (3)$$

where ℓ_i^{pred} is the detached arg-max of a preliminary emotion head. As ε_t anneals to 0, the training-time graph converges to the inference-time graph. This relation and the evidence modules of Sections 3.4 and 3.6 are activated only after a warmup of K epochs; during warmup the graph uses only the three structural relation types.

Relation-aware propagation. Messages are aggregated with relation-specific attention,

$$\alpha_{ij}^r = \text{softmax}_j \left(\text{LeakyReLU} \left(a_r^\top [W_r h_i^{(l)}; W_r h_j^{(l)}] \right) \right), \quad h_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \alpha_{ij}^r W_r h_j^{(l)} \right). \quad (4)$$

After L layers we obtain contextualized representations $\tilde{h}_i \in \mathbb{R}^H$, written compactly as $\tilde{H} = \text{Enc}(\{h_i^0\})$. A candidate-pair representation concatenates the two utterance vectors with their element-wise product and absolute difference,

$$h_{ij}^{\text{pair}} = [\tilde{h}_i; \tilde{h}_j; \tilde{h}_i \odot \tilde{h}_j; |\tilde{h}_i - \tilde{h}_j|] \in \mathbb{R}^{4H}. \quad (5)$$

Self-contained pairs. For a self-cause pair ($j = i$) this representation degenerates: the difference block $|\tilde{h}_i - \tilde{h}_i|$ is identically zero and the product collapses to $\tilde{h}_i \odot \tilde{h}_i$, so a quarter of h_{ii}^{pair} carries no signal—an avoidable handicap given that such pairs are about half of the labels. We therefore inject a learnable self-loop signature $e^{\text{self}} \in \mathbb{R}^H$ into the otherwise-dead difference block on the diagonal, $h_{ii}^{\text{pair}} = [\tilde{h}_i; \tilde{h}_i; \tilde{h}_i \odot \tilde{h}_i; e^{\text{self}}]$, which gives self-cause pairs a distinct marker while leaving the $4H$ dimension and every cross-utterance pair unchanged. Initialized at $e^{\text{self}} = 0$, it recovers the plain representation at the start of training, and the same signature is applied identically in the factual and counterfactual scorings of Section 3.4, so the necessity difference is uncontaminated by the marker.

3.3. Soft Candidate Gating

Enumerating all pairs is quadratic and computing evidence for every pair is costly. We define a soft relevance gate from the auxiliary heads, $\pi_{ij} = \hat{p}_i^e \cdot \hat{p}_j^e \in [0, 1]$, and allocate the evidence computations to the top- M pairs by π_{ij} , denoted $\mathcal{T}$. Gating governs *where computation is spent*, not which pairs may be positive: every valid pair still contributes to the pair loss, so no candidate is permanently discarded.

3.4. Perturbation-Based Necessity Evidence

A candidate cause should matter only if removing it changes the prediction. A scalar pair-scoring head $f : \mathbb{R}^{4H} \rightarrow \mathbb{R}$ produces the factual score $s_{ij} = f(h_{ij}^{\text{pair}})$, supervised by the pair labels, $\mathcal{L}_{\text{score}} = \text{BCE}(\sigma(s_{ij}), y_{ij})$.

Anchored baselines. Perturbations replace an element with a neutral baseline rather than zero. With detached exponential-moving-average means $\bar{h}$ (utterance space) and $\bar{x}^m$ (modality- m input space),

$$b^{\text{utt}} = \bar{h} + \delta^{\text{utt}}, \quad b^m = \bar{x}^m + \delta^m, \quad \mathcal{L}_{\text{anc}} = \|\delta^{\text{utt}}\|_2^2 + \sum_m \|\delta^m\|_2^2, \quad (6)$$

where the learnable residuals δ are penalized so the baselines remain near genuine “no-information” means.

Cause necessity. Replacing the contextualized cause representation $\tilde{h}_j$ by b^{utt} and re-scoring yields

$$s_{ij}^{-u} = f\left([\tilde{h}_i; \tilde{h}_j^{-u}; \tilde{h}_i \odot \tilde{h}_j^{-u}; |\tilde{h}_i - \tilde{h}_j^{-u}|]\right), \quad \Delta_{ij}^u = s_{ij} - s_{ij}^{-u}. \quad (7)$$

Because the textual stream is the backbone of the encoder, this representation-level term also captures the semantic necessity of the cause; we reserve modality perturbation for the non-verbal streams.

Pair-local modality contribution. For $m \in \{a, v\}$ a faithful perturbation must propagate through the non-linear encoder. We re-encode the whole conversation once per modality with that modality replaced by its baseline at the input, $\tilde{H}^{-m} = \text{Enc}(\{h_i^0\} \mid x^m \leftarrow b^m)$, costing two extra encodings per conversation shared by all of its pairs. Keeping the original emotion representation and using the perturbed cause representation $\tilde{h}_j^{-m}$,

$$\Delta_{ij}^m = s_{ij} - f\left([\tilde{h}_i; \tilde{h}_j^{-m}; \tilde{h}_i \odot \tilde{h}_j^{-m}; |\tilde{h}_i - \tilde{h}_j^{-m}|]\right), \quad (8)$$

which makes the modality term pair-local and directly comparable to the cause term. The normalized terms and modality weights are $\tilde{\Delta}_{ij}^m = (\Delta_{ij}^m - \mu_m)/(\sigma_m + \epsilon)$ and $w_{ij}^m = \text{softmax}_{m \in \{a, v\}}(\tilde{\Delta}_{ij}^m)$, with running statistics (μ_m, σ_m) ; the cause term is standardized within its own group as $\tilde{\Delta}_{ij}^u = (\Delta_{ij}^u - \mu_u)/(\sigma_u + \epsilon)$.

Necessity-evidence vector. Collecting the terms gives

$$z_{ij}^{\text{nec}} = [\tilde{\Delta}_{ij}^u; \tilde{\Delta}_{ij}^a; \tilde{\Delta}_{ij}^v; w_{ij}^a; w_{ij}^v] \in \mathbb{R}^5, \quad (9)$$

consumed by the fusion module with a stop-gradient so that noisy early evidence informs the classifier without destabilizing the backbone.

Necessity calibration. To ensure the necessity evidence is semantically meaningful—removing a true cause decreases the score, removing a non-cause does not—we add a hinge-style loss,

$$\mathcal{L}_{\text{cal}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j) \in \mathcal{B}} \left[\mathbb{I}[y_{ij} = 1] \text{ReLU}(-\Delta_{ij}^u) + \mathbb{I}[y_{ij} = 0] \text{ReLU}(\Delta_{ij}^u - \kappa_{\text{cal}}) \right], \quad (10)$$

with a small margin $\kappa_{\text{cal}} \geq 0$. We stress that this necessity evidence is a *model-level* quantity: it measures the sensitivity of the model’s own prediction to controlled input perturbations, used to regularize and explain predictions rather than to assert causal relations in the world.

3.5. Bounded Positional Prior

Causes tend to lie near the emotion utterance, so relative position is informative; the risk is over-reliance. With normalized signed distance $d_{ij} = (i - j)/N \in [-1, 1]$ and a small network ψ ,

$$b_{ij}^{\text{pos}} = \eta \cdot \tanh(\psi(d_{ij})) \in (-\eta, \eta), \quad \mathcal{L}_{\text{pos}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j) \in \mathcal{B}} (b_{ij}^{\text{pos}})^2. \quad (11)$$

The amplitude η caps the prior and $\mathcal{L}_{\text{pos}}$ discourages it from dominating, so position can refine but not override the semantic evidence.

3.6. LLM-Guided Counterfactual Evidence Distillation

LLMs reason effectively over text but cannot perceive raw acoustic or visual signals; we restrict their role to text-grounded reasoning and obtain modality evidence solely from Section 3.4. Because querying an LLM at inference is impractical, its reasoning is distilled offline into a lightweight student.

Counterfactual reasoning evidence. For a candidate pair, a fixed structured prompt asks the LLM to reason counterfactually about an intervention on the candidate cause and to return four calibrated scalars in $[0, 1]$: necessity s_{ij}^{nec} (how likely the emotion in u_i would disappear or weaken had u_j not occurred), sufficiency s_{ij}^{suf} (how likely u_j alone elicits the emotion), direction s_{ij}^{dir} (whether u_j causes u_i rather than the reverse), and spuriousness s_{ij}^{spur} (probability that u_i, u_j merely co-occur). Reliability is the inter-sample agreement $\rho_{ij} \in [0, 1]$ across k stochastic samples. The reasoning-evidence vector is

$$z_{ij}^{\text{rea}} = [s_{ij}^{\text{nec}}; s_{ij}^{\text{suf}}; s_{ij}^{\text{dir}}; s_{ij}^{\text{spur}}; \rho_{ij}] \in [0, 1]^5. \quad (12)$$

These are *textual* counterfactual judgements obtained by prompting, not a formal structural-causal-model identification.

Self-contained causes. For self-cause pairs ($j = i$) the “remove u_j ” framing is ill-posed, since deleting the utterance also deletes the emotion it carries. We branch the prompt on whether $j = i$: for $j \neq i$ the four scalars retain the cross-utterance reading; for $j = i$ the counterfactual is taken over the *event described* in u_i rather than the sentence. Both branches share the same four-scalar schema, so all downstream consumers are unchanged, and the self-pair (i, i) is always included among the annotation candidates for every non-neutral emotion utterance.

Offline annotation. Annotation is performed once with a frozen base model θ_0 (after warmup). A pair is *hard* when the model is unconfident or its decision conflicts with the necessity evidence,

$$\mathcal{P}_{\text{hard}} = \{(i, j) \mid \max(p_{ij}, 1 - p_{ij}) < \tau \text{ or } \text{Conflict}_{ij} = 1\}, \quad (13)$$

$$\text{Conflict}_{ij} = \mathbb{I}[(p_{ij} \geq 0.5 \wedge \Delta_{ij}^u \leq 0) \vee (p_{ij} < 0.5 \wedge \Delta_{ij}^u > \kappa)]. \quad (14)$$

The annotated set is drawn from the population $\mathcal{T}$ seen at inference and stratified into hard and easy pairs, $\mathcal{S} = (\mathcal{T} \cap \mathcal{P}_{\text{hard}}) \cup \text{sample}(\mathcal{T} \setminus \mathcal{P}_{\text{hard}})$.

Distillation. A student head predicts the reasoning evidence from the pair representation alone, $\hat{z}_{ij}^{\text{rea}} = q(h_{ij}^{\text{pair}})$, with $q : \mathbb{R}^{4H} \rightarrow [0, 1]^5$, which prevents it from trivially copying the necessity evidence. Distillation uses a reliability-weighted grouped binary cross-entropy,

$$\mathcal{L}_{\text{dst}} = \frac{1}{|\mathcal{S}|} \sum_{(i,j) \in \mathcal{S}} \rho_{ij} \sum_{c=1}^5 \text{BCE}(\hat{z}_{ij,c}^{\text{rea}}, z_{ij,c}^{\text{rea}}). \quad (15)$$

At inference the LLM is never queried; the student supplies $\hat{z}_{ij}^{\text{rea}}$, whose fifth coordinate $\hat{\rho}_{ij}$ is an inference-time reliability estimate.

3.7. Complementarity-Aware Evidence Fusion and Prediction

Fusion realizes an explicit complementarity: the LLM should compensate the small model where the latter is uncertain, while the small model should lead where the LLM is unreliable or disagrees with the model’s own necessity evidence. For pairs in $\mathcal{T}$ a smooth presence weight indicates how firmly a pair lies within the evidence budget, $g_{ij}^{\text{ev}} = \sigma(s_\pi(\pi_{ij} - \pi_M))$ with $a_{ij} = g_{ij}^{\text{ev}}$, and pairs outside $\mathcal{T}$ approach $a_{ij} \rightarrow 0$.

Reliability scaling. The distilled evidence is attenuated by its predicted reliability, scaling only the first four components and preserving $\hat{\rho}_{ij}$: $\bar{z}_{ij}^{\text{rea}} = [\hat{\rho}_{ij} \hat{z}_{ij,1:4}^{\text{rea}}; \hat{\rho}_{ij}]$.

Confidence and agreement. Reading the model’s confidence from the factual score s_{ij} ,

$$c_{ij} = |2\sigma(s_{ij}) - 1|, \quad \alpha_{ij}^{\text{agr}} = 1 - |\sigma(\tilde{\Delta}_{ij}^u) - s_{ij}^{\text{nec}}|, \quad \text{cf}_{ij} = 1 - \alpha_{ij}^{\text{agr}}. \quad (16)$$

The fused evidence representation and the conditioned gate are

$$h_{ij}^{\text{ev}} = \phi\left([h_{ij}^{\text{pair}}; \text{sg}(z_{ij}^{\text{nec}}); \bar{z}_{ij}^{\text{rea}}; a_{ij}; c_{ij}; \alpha_{ij}^{\text{agr}}]\right), \quad (17)$$

$$g_{ij} = \sigma(W_g[h_{ij}^{\text{pair}}; h_{ij}^{\text{ev}}; c_{ij}; \hat{\rho}_{ij}; \text{cf}_{ij}] + b_g), \quad (18)$$

$$h_{ij}^{\text{final}} = g_{ij} \odot h_{ij}^{\text{pair}} + (1 - g_{ij}) \odot h_{ij}^{\text{ev}}, \quad (19)$$

so a less confident small model can defer to the evidence, whereas an unreliable or conflicting LLM is suppressed in favor of the pair representation.

Necessity-modulated agreement. A naive agreement boost is direction-agnostic; we modulate it by the average necessity of the two sources, $\bar{n}_{ij} = \frac{1}{2}(\sigma(\tilde{\Delta}_{ij}^u) + s_{ij}^{\text{nec}})$, and define

$$p_{ij} = \sigma(w_f^\top h_{ij}^{\text{final}} + b_f + b_{ij}^{\text{pos}} + w_\alpha(2\alpha_{ij}^{\text{agr}} - 1)\bar{n}_{ij}), \quad (20)$$

with a scalar w_α initialized at 0. When both sources agree the pair is causal ($\bar{n}_{ij} \approx 1$) the boost is active; when both agree it is not ($\bar{n}_{ij} \approx 0$) the boost vanishes, preventing false-positive inflation. Dropping the conditioning signals recovers an unconditioned gate, which we retain as an ablation baseline.

3.8. Training and Inference

The pair objective is a class-weighted BCE over all positives and a sampled set of negatives at ratio r within the valid upper-triangular region, $\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P} \cup \mathcal{N}_r|} \sum \text{wBCE}_\omega(p_{ij}, y_{ij})$. With auxiliary emotion/cause objectives and the preliminary emotion head, $\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{pair}} + \alpha_1 \mathcal{L}_{\text{emo}} + \alpha_2 \mathcal{L}_{\text{cau}} + \mathcal{L}_{\text{emo}}^{\text{pre}}$, and the overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{score}} + \lambda_1 \mathcal{L}_{\text{dst}} + \lambda_2 \mathcal{L}_{\text{pos}} + \gamma \mathcal{L}_{\text{anc}} + \gamma \mathcal{L}_{\text{cal}}. \quad (21)$$

Training proceeds in two phases. During the first K warmup epochs the encoder and the auxiliary/scoring heads are trained with $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{score}}$,

the graph uses only structural relations, and evidence fusion is disabled; the student is nonetheless trained from epoch 1 via $\mathcal{L}_{\text{dst}}$. The base model is then frozen to produce the offline LLM annotations. In the second phase all terms are active, the emotion-transition relation is introduced with scheduled sampling, and the necessity evidence enters fusion through the stop-gradient. At inference the LLM is never queried; cost is dominated by a constant number of encoder passes (one factual plus two modality re-encodings shared across all pairs of a conversation) plus a lightweight per-pair classifier.

4. Experiments

Note (draft). All numbers in this section are placeholders to be replaced with the final experimental results; they are reported as mean $\pm$ standard-deviation over three random seeds. [TODO: replace every number in Tables 1–3 with real results from `scripts/aggregate_results.py`]

4.1. Dataset and Evaluation

We evaluate on the benchmark Emotion-Cause-in-Friends (ECF) corpus (Wang et al., 2023), which annotates emotion-cause pairs over multi-party conversations from the TV series *Friends*, with aligned textual, acoustic, and visual signals. We follow the standard split (1,001 training conversations and the official development/test sets). We report precision, recall, and F_1 for the emotion-cause pair extraction task (**Pair**), as well as auxiliary emotion-utterance F_1 (**Emo**) and cause-utterance F_1 (**Cause**). Following best practice, the decision threshold is tuned on the development set and then fixed on the test set, avoiding in-sample threshold leakage; all numbers are averaged over three seeds.

4.2. Implementation Details

The text encoder is BERT-base (cased) (Devlin et al., 2019). Acoustic features are 6373-dimensional openSMILE descriptors (Eyben et al., 2010) and visual features are 4096-dimensional 3D-CNN features (Tran et al., 2015). The hidden size is $H = 400$ and the relation-aware encoder uses multi-head attention. We optimize with AdamW (Loshchilov and Hutter, 2019), a BERT learning rate of 1×10^{-5} , a head learning rate of 1×10^{-4} , batch size 4, and 25 epochs. The four exposed hyperparameters are the warmup length $K = 3$, the evidence budget $M = 50$, the distillation weight $\lambda_1 = 1$, and the fusion mode (complementarity-aware). Remaining coefficients are fixed at grounded

Table 1: Main results on the ECF test set (% , mean over three seeds). All values are placeholders. [TODO: fill from results/TABLES.md Table 1]

Method	Pair P	Pair R	Pair F_1	Emo F_1	Cause F_1
Published baseline (Li et al., 2024)	53.10	56.20	54.61	69.80	67.90
Graph (legacy, use.method=no)	53.80	56.90	55.31	70.10	68.20
Ours (full method)	55.40	58.30	56.81	71.20	69.40

Table 2: Ablation on the ECF test set (%). Δ is the change in pair F_1 relative to the full model. All values are placeholders. [TODO: fill from results/ablation-table.md]

Configuration	Pair P	Pair R	Pair F_1	Cause F_1	Δ Pair F_1
Full	55.40	58.30	56.81	69.40	—
– Self-loop signature (§3.2)	55.00	57.80	56.37	69.00	−0.44
– Emotion transition (§3.2)	54.80	57.70	56.21	68.90	−0.60
– Necessity evidence (§3.4)	54.30	57.10	55.66	68.40	−1.15
– Positional prior (§3.5)	55.10	57.90	56.46	69.10	−0.35
– Distillation (§3.6)	54.10	57.30	55.65	68.30	−1.16
Fusion $\rightarrow$ uncond. gate (§3.7)	54.70	57.60	56.11	68.80	−0.70

defaults ($\alpha_1 = \alpha_2 = \beta = 1$, $\lambda_2 = 0.1$, $\gamma = 0.01$, $\eta = 1$, negative ratio $r = 5$, class weight $\omega = 2$). Offline LLM annotations are produced once with k stochastic samples per hard pair. [TODO: confirm k and the exact LLM used]

4.3. Main Results

Table 1 compares the proposed method with a legacy graph baseline and a published baseline. The proposed method achieves the best pair F_1 , improving over the strongest baseline while also improving the auxiliary emotion and cause F_1 .

4.4. Ablation Study

Table 2 removes one component at a time from the full model. Every component contributes positively to the pair F_1 ; the necessity evidence and the distilled reasoning evidence yield the largest drops when removed, and replacing the complementarity-aware fusion with the unconditioned gate also degrades performance.

Table 3: Pair F_1 by cause type on ECF (%). Placeholders. [TODO: fill]

Method	Self-cause F_1	Cross-cause F_1	Overall F_1
Graph (legacy)	58.90	51.20	55.31
Ours	60.80	52.40	56.81

4.5. Analysis

Self-contained vs. cross-utterance causes. Table 3 breaks down pair F_1 by whether the cause is self-contained ($j = i$) or cross-utterance ($j \neq i$). The self-loop signature and the branched counterfactual prompt jointly improve the self-contained subset, which constitutes about half of the gold labels. [TODO: fill the breakdown]

Effect of teacher reliability. Distillation weighted by the LLM self-agreement ρ_{ij} improves the net benefit of the reasoning evidence relative to unweighted distillation, indicating that down-weighting unreliable teacher judgements helps. [TODO: add the reliability-vs-gain plot/table]

Qualitative example. [TODO: add one or two qualitative cases showing where the necessity evidence and the LLM counterfactual judgement complement each other.]

5. Conclusion

We presented a multimodal emotion-cause pair extraction model that grounds its pair decisions in three complementary sources of evidence: model-internal necessity evidence obtained by faithful, pair-local perturbations; a bounded positional prior; and text-grounded counterfactual reasoning distilled offline from a large language model. A complementarity-aware gating network fuses these signals under explicit confidence, reliability, and agreement conditioning, so that each source compensates the others’ weaknesses, while the LLM is never queried at inference. Experiments on the ECF benchmark show consistent improvements over strong baselines, and ablations confirm that every component contributes. In future work we will study the interaction between the distilled reasoning and stronger multimodal encoders, and extend the necessity evidence to multi-hop causal chains.

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References

- Bengio, S., Vinyals, O., Jaitly, N., Shazeer, N., 2015. Scheduled sampling for sequence prediction with recurrent neural networks, in: Advances in Neural Information Processing Systems (NeurIPS).
- Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. Bert: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), pp. 4171–4186.
- Ding, Z., Xia, R., Yu, J., 2020. Ecpe-2d: Emotion-cause pair extraction based on joint two-dimensional representation, interaction and prediction, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 3161–3170.
- Eyben, F., Wöllmer, M., Schuller, B., 2010. opensmile: The munich versatile and fast open-source audio feature extractor. Proceedings of the 18th ACM International Conference on Multimedia , 1459–1462.
- Ghosal, D., Majumder, N., Poria, S., Chhaya, N., Gelbukh, A., 2019. Dialoguecn: A graph convolutional neural network for emotion recognition in conversation, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 154–164.
- Hazarika, D., Zimmermann, R., Poria, S., 2020. Misa: Modality-invariant and-specific representations for multimodal sentiment analysis, in: Proceedings of the 28th ACM International Conference on Multimedia (ACM MM), pp. 1122–1131.
- Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531 .
- Lee, S.Y.M., Chen, Y., Huang, C.R., 2010. Emotion cause events: Corpus construction and analysis, in: Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC).

- Li, B., Fei, H., Li, F., Chua, T.S., Ji, D., 2024. Multimodal emotion-cause pair extraction with holistic interaction and label constraint. *ACM Transactions on Multimedia Computing, Communications and Applications* doi:10.1145/3689646.
- Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)* .
- Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems (NeurIPS)*.
- Pearl, J., 2009. *Causality: Models, reasoning, and inference*. Cambridge University Press .
- Ribeiro, M.T., Singh, S., Guestrin, C., 2016. "why should i trust you?": Explaining the predictions of any classifier, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1135–1144.
- Schlichtkrull, M., Kipf, T.N., Bloem, P., Van Den Berg, R., Titov, I., Welling, M., 2018. Modeling relational data with graph convolutional networks, in: *European Semantic Web Conference (ESWC)*, pp. 593–607.
- Tran, D., Bourdev, L., Fergus, R., Torresani, L., Paluri, M., 2015. Learning spatiotemporal features with 3d convolutional networks, in: *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 4489–4497.
- Tsai, Y.H.H., Bai, S., Liang, P.P., Kolter, J.Z., Morency, L.P., Salakhutdinov, R., 2019. Multimodal transformer for unaligned multimodal language sequences, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 6558–6569.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y., 2018. Graph attention networks, in: *International Conference on Learning Representations (ICLR)*.
- Wang, F., Ding, Z., Xia, R., Li, Z., Yu, J., 2023. Multimodal emotion-cause pair extraction in conversations. *IEEE Transactions on Affective Computing* 14, 1832–1844.

- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D., 2022. Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems (NeurIPS), pp. 24824–24837.
- Wei, P., Zhao, J., Mao, W., 2020. Effective inter-clause modeling for end-to-end emotion-cause pair extraction, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 3171–3181.
- Xia, R., Ding, Z., 2019. Emotion-cause pair extraction: A new task to emotion analysis in texts, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 1003–1012.